Arjun Manoj | CEC23CS043 | CSQ413 Seminar

- Paper name: Real-Time Detection Method for Cracked Eggs Based on Improved YOLOv8 and Dual Verification With Triple Classification Granularity.
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- This paper basically solves the matter of detecting cracked eggs on industrial production lines (like in a poultry conveyor belt) in real-time at very high speeds with very high precision using a customized algorithm.
- It is based on the improved YOLOv8 framework. YOLOv8 is a fast and accurate computer vision model built by Ultralytics for tasks like object detection, image classification and image segmentation. YOLO stands for "You Only Look Once". It is a popular group of algorithms. Instead of scanning an image in multiple steps, YOLO processes the whole image in one single pass. This makes it very fast and ideal for real-time video analysis such as in the case of cracked egg detection.

- About 3% of eggs get damaged during transport and production. Currently, many factories rely on human manual inspection, which is very slow and inaccurate. Some automatic methods use sound (acoustic methods), but they are easily disturbed by factory noise and are expensive. Older computer vision methods struggle to keep up with the high speed of real production lines (they need to process at least 30 frames per second or 30 FPS). Existing models also fail to detect cracks on the bottom of the egg because they do not use a rotating mechanism to see the whole shell.
- Instead of a simple good egg vs bad egg (binary) classification, this paper uses three categories: intact eggs, cracked eggs and crack regions. The crack region specifically targets the localized damaged area. The system uses a dual verification rule: an egg is marked as broken if the AI detects either a cracked egg overall or a specific crack region. This drastically reduces the chances of missing a broken egg. The physical conveyor belt also rotates the eggs as they move, ensuring the camera sees all sides so that cracks on the bottom of the egg can also be detected.
- Technologies used and their purpose:
    - Deformable Convolutional Networks v2 (DCNv2): Standard image convolutions use rigid, square grids. DCNv2 allows the network to dynamically change its shape to match the irregular, jagged lines of eggshell cracks.
    - BiFPN (Weighted Bidirectional Feature Pyramid Network): This replaces the standard PANet in YOLOv8. It efficiently mixes low-level visual details (like tiny textures) with high-level meanings (the shape of the egg), helping the model spot very subtle cracks.
    - WIoUv2 Loss Function: This updated mathematical loss function helps the AI balance its learning. It forces the network to pay more attention to hard-to-detect samples (like blurry or heavily hidden cracks) during training.
    - Small Object Detection Layer: The default YOLOv8 model shrinks images in a way that makes tiny objects disappear. The authors added a dedicated layer specifically to catch very small targets, which is perfect for microscopic hairline cracks.
- The dataset was custom-made by capturing real, unwashed eggs on a moving production line at Guangzhou Suiping Enterprise Co., Ltd. This ensured the AI learned from real-world factory lighting, backgrounds and

motion. This dataset contained 628 images initially before processing/augmentation. After applying data augmentation (like flipping and cropping images to give the AI more examples to learn from), the final dataset contained 3,222 images. These images contained 7,611 intact eggs, 2,711 cracked eggs and 2,935 cracked areas.

- Evaluation metrics used and the results of the proposed system:
    - Precision (P): The percentage of eggs the model called cracked that were actually cracked. It is 94.1% for this proposed algorithm.
    - Recall (R): The percentage of actual cracked eggs that the model successfully found. It is 90% for this proposed algorithm.
    - mAP (mean Average Precision): A comprehensive score combining both Precision and Recall to judge overall accuracy. The improved YOLOv8 model achieved an mAP of 92.9% on the test dataset. This is a 5.2% improvement over the standard, unmodified YOLOv8 model. It easily beat other popular models like Faster R-CNN, YOLOv5, and even YOLOv11 in accuracy.
    - FPS (Frames Per Second): How fast the model can process live video. In live factory tests, the model ran at 45 FPS. While slightly slower than the original YOLOv8 (52 FPS) due to the added complex layers, it is still well above the 30 FPS needed for real-time industrial use.
    - FNR (False Negative Rate) and FPR (False Positive Rate): Used in the factory testing to see how many broken eggs slipped through (FNR) and how many good eggs were accidentally thrown away (FPR). The false negative rate for broken eggs dropped to just 1.06%, meaning almost zero broken eggs made it past the camera.